# EE 121B: Principles of Semiconductor Device Design

**Lecture 4:**

**Semiconductor Fundamentals (continued)**

Sam Emaminejad

# Lecture 4

OUTLINE

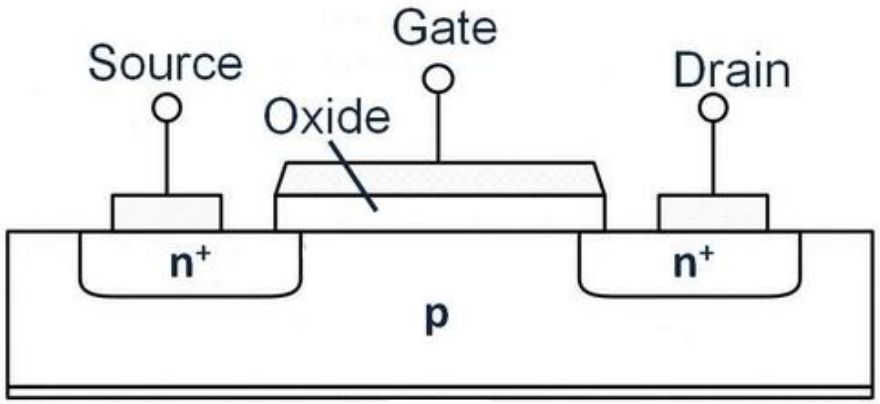
- Semiconductor Fundamentals (cont'd)
  - Scattering Mechanisms
  - Carrier actions:
    - Drift
    - Diffusion
    - Generation-Recombination

  Reading: Pierret 3.2-3.3; Hu 2.3, 2.5-2.6

# Course Overview

1. Semiconductor Properties
2. P-N Junction
3. Metal-Semiconductor Contacts
4. MOS Devices
5. Bipolar Junction Transistor

**Metal-Oxide-Semiconductor (MOS)**
**Field-Effect Transistor (FET)**




http://qph.is.quoracdn.net/main-qimg-38a99b4f215e161bf31964795cfce7d2?convert_to_webp=true

# Mechanisms of Carrier Scattering

Dominant scattering mechanisms:

1. Phonon scattering (lattice scattering)
2. Impurity (dopant) ion scattering

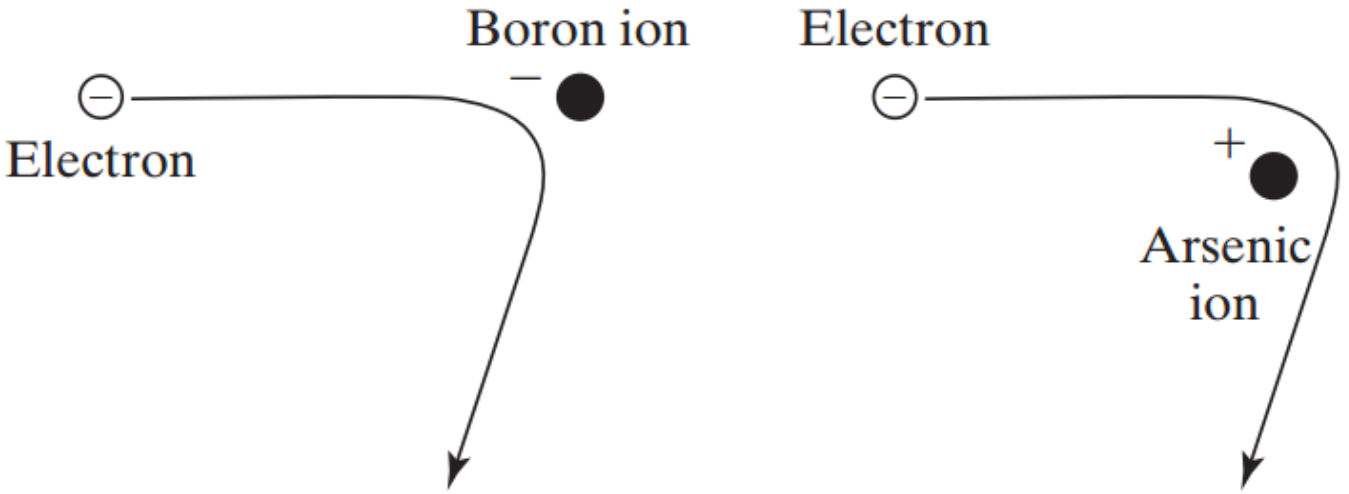
**Phonon scattering limited mobility decreases with increasing *T*:**

$$\mu_{phonon} \propto \tau_{phonon} \propto \frac{1}{\text{phonon density} \times \text{carrier thermal velocity}} \propto \frac{1}{T \times T^{1/2}} \propto T^{-3/2}$$

$\mu = q\tau / m$

$v_{th} \propto \sqrt{T}$

# Impurity Ion Scattering




There is less change in the electron's direction if the electron travels by the ion at a higher speed.

**Ion scattering limited mobility** increases **with increasing *T*:**

$$\mu_{impurity} \propto \frac{v_{th}^3}{N_A + N_D} \propto \frac{T^{3/2}}{N_A + N_D}$$

# Matthiessen's Rule

- The probability that a carrier will be scattered by mechanism *i* within a time period *dt* is $dt/\tau_i$

$\tau_i$ ≡ mean time between scattering events due to mechanism *i*

→ Probability that a carrier will be scattered by any mechanism within a time period *dt* is $\sum_i dt/\tau_i$

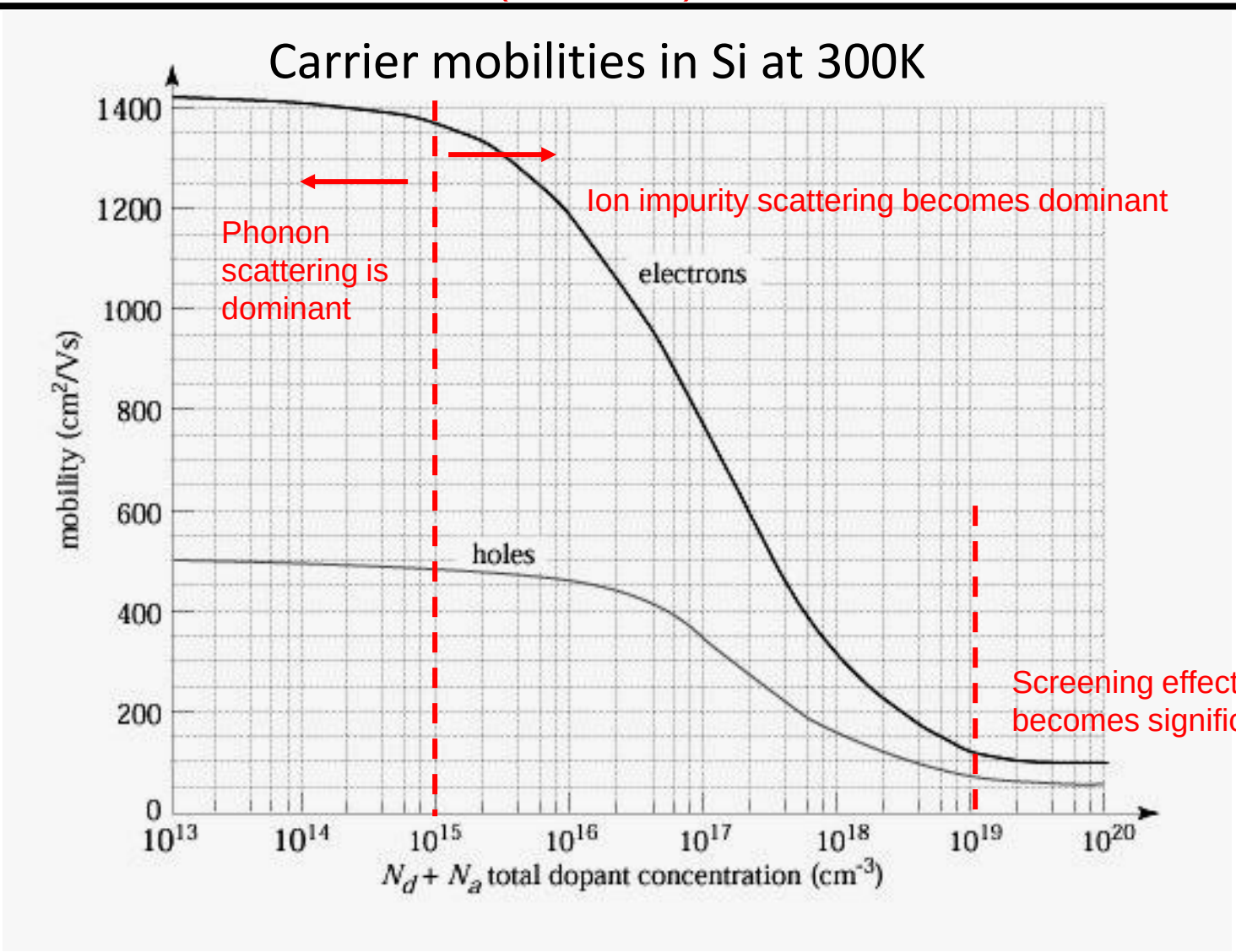
$$\mu = \frac{q\tau}{m^*}$$

$$\frac{1}{\tau} = \frac{1}{\tau_{phonon}} + \frac{1}{\tau_{impurity}} \quad \Rightarrow \quad \frac{1}{\mu} = \frac{1}{\mu_{phonon}} + \frac{1}{\mu_{impurity}}$$

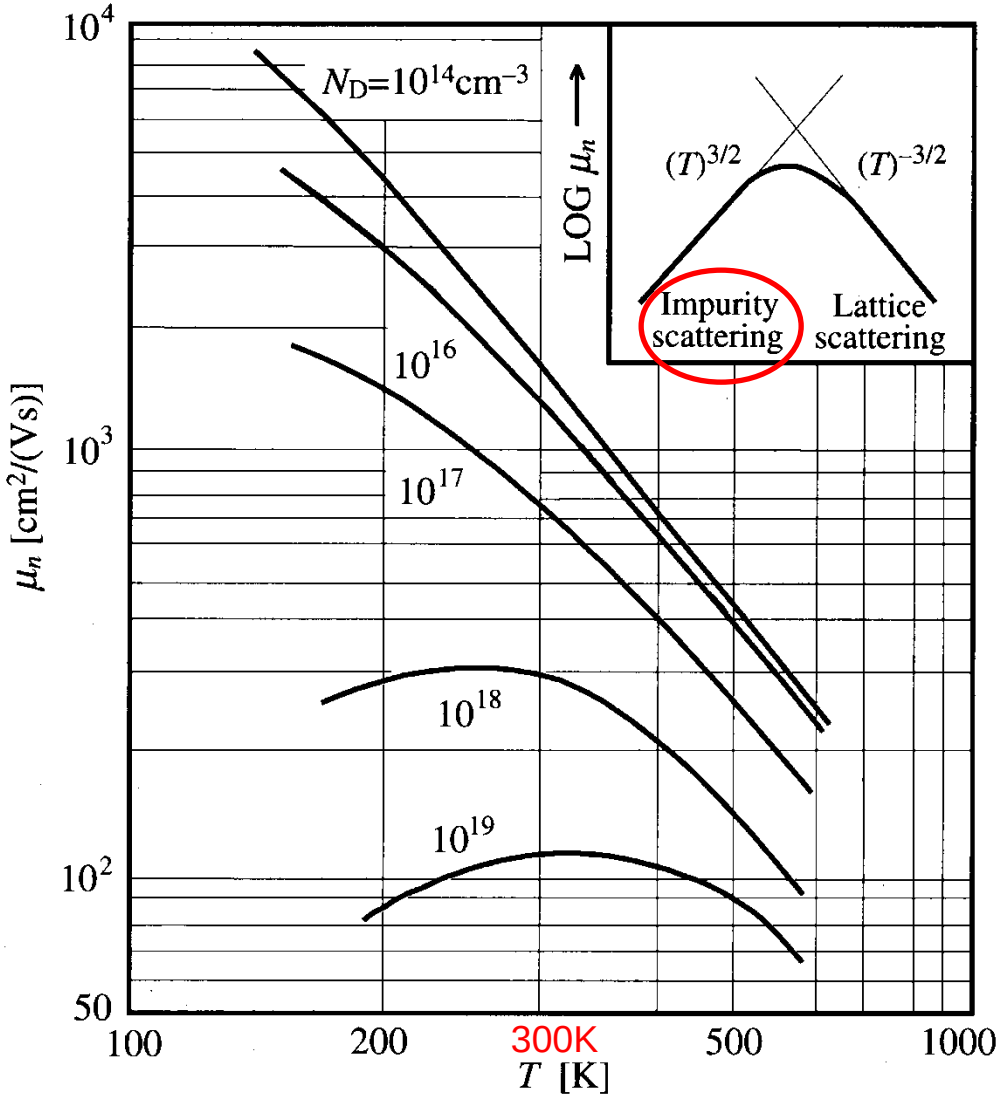
The dominant scattering mechanism (with the lowest associated $\mu_i$) determines $\mu$

# Mobility Dependence on Doping

$(T = 300K)$

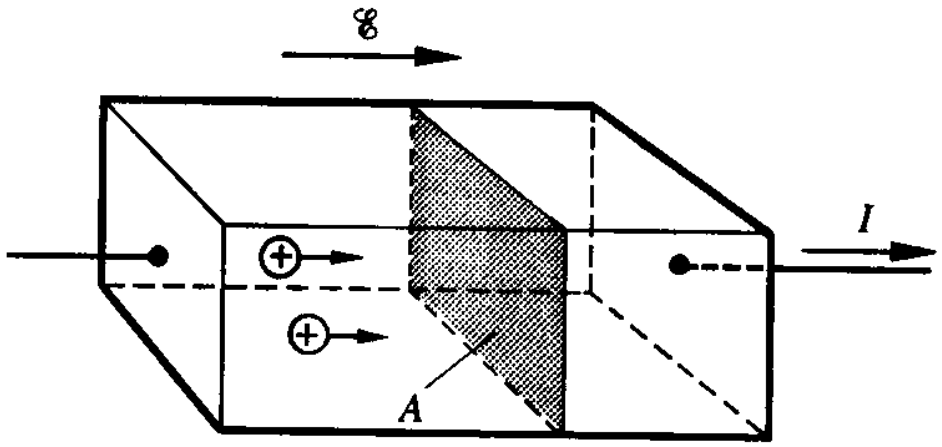


# Mobility Dependence on Temperature




$$\frac{1}{\mu} = \frac{1}{\mu_{phonon}} + \frac{1}{\mu_{impurity}}$$

# Hole Drift Current Density, $J_{p,drift}$




R.F. Pierret, *Semiconductor Fundamentals*, Figure 3.3

$v_{dp}\,\Delta t\,A$ = volume from which all holes cross plane in time $\Delta t$

$p\,v_{dp}\,\Delta t$ A = number of holes crossing plane in time $\Delta t$

$q\,p\,v_{dp}\,\Delta t$ A = hole charge crossing plane in time $\Delta t$

$q\,p\,v_{dp}\,A$ = hole charge crossing plane per unit time = hole current

→ **Hole drift current per unit area $J_{p,drift} = q\,p\,v_{dp}$**

# Conductivity and Resistivity

- In a semiconductor, both electrons and holes conduct current:

$$J_{p,drift} = qp\mu_p \mathcal{E} \qquad J_{n,drift} = -qn(-\mu_n \mathcal{E})$$

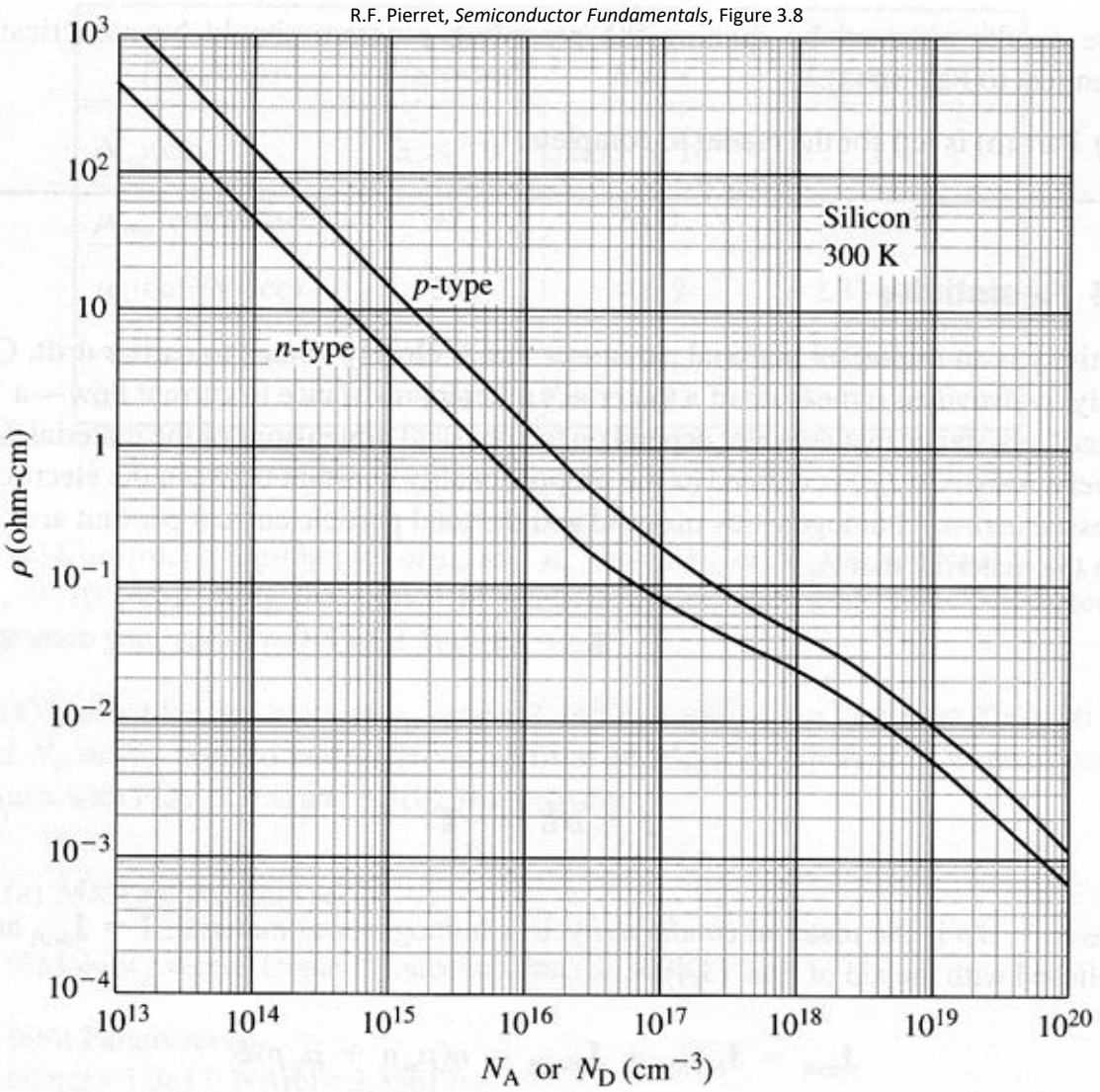
$$J_{drift} = J_{p,drift} + J_{n,drift} = qp\mu_p \mathcal{E} + qn\mu_n \mathcal{E}$$

$$J_{drift} = (qp\mu_p + qn\mu_n)\mathcal{E} \equiv \sigma \mathcal{E} \qquad \sigma = \frac{J}{\mathcal{E}}$$

- The ***conductivity*** of a semiconductor is $\boxed{\sigma \equiv qp\mu_p + qn\mu_n}$
  - Unit: mho/cm

- The ***resistivity*** of a semiconductor is $\boxed{\rho \equiv \frac{1}{\sigma}}$
  - Unit: ohm-cm

# Resistivity Dependence on Doping




**For n-type material:**

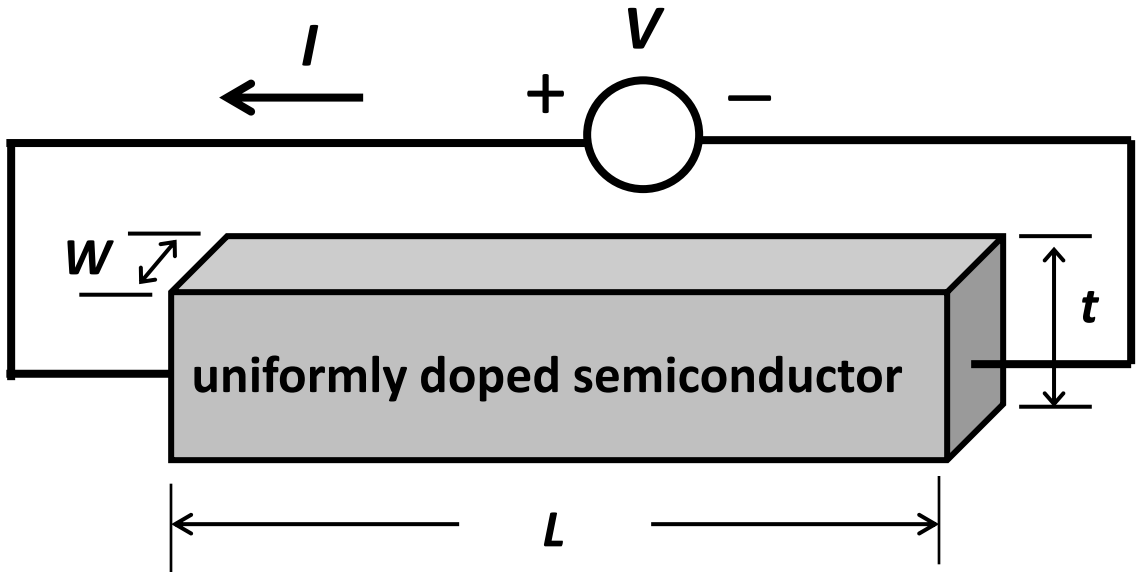
$$\rho \cong \frac{1}{qn\mu_n}$$

**For p-type material:**

$$\rho \cong \frac{1}{qp\mu_p}$$

**Note: This plot (for Si) does not apply to compensated material (doped with both acceptors and donors).**

# Electrical Resistance




**Resistance**

$$R \equiv \frac{V}{I} = \rho \frac{L}{Wt}$$

**[Unit: ohms]**

where $\rho$ is the resistivity

# Example: Resistivity Calculation

What is the resistivity of a Si sample doped with $10^{16}/cm^3$ Boron?

Answer:

$N_A = 10^{16}\text{cm}^{-3},\ N_D = 0$

$p = 10^{16}\text{cm}^{-3},\ n = 10^4 cm^{-3}$

$n_1 = n_i^2 = 10^{20}\text{cm}^{-6}$

$$\rho = \frac{1}{qn\mu_n + qp\mu_p} \cong \frac{1}{qp\mu_p}$$

$$= \left[(1.6\times10^{-19})(10^{16})(450)\right]^{-1} = 1.4\,\Omega - \text{cm}$$

# Example: Compensated Doping

Consider the same Si sample doped with $10^{16}/cm^3$ Boron, and *additionally* doped with $10^{17}/cm^3$ Arsenic. What is its resistivity?

Answer:

$N_D = 10^{17}\text{cm}^{-3}, N_A = 10^{16}\text{cm}^{-3}$
$n = 9 \times 10^{16} cm^{-3}$

$$\rho = \frac{1}{qn\mu_n + qp\mu_p} \cong \frac{1}{qn\mu_n}$$

$$= \left[(1.6\times10^{-19})(9\times10^{16})(750)\right]^{-1} = 0.93\,\Omega - \text{cm}$$

# Example: *T* Dependence of $\rho$

Consider a Si sample doped with $10^{17}$ As atoms/cm$^3$. How will its resistivity change when *T* is increased from 300K to 400K?

Answer:

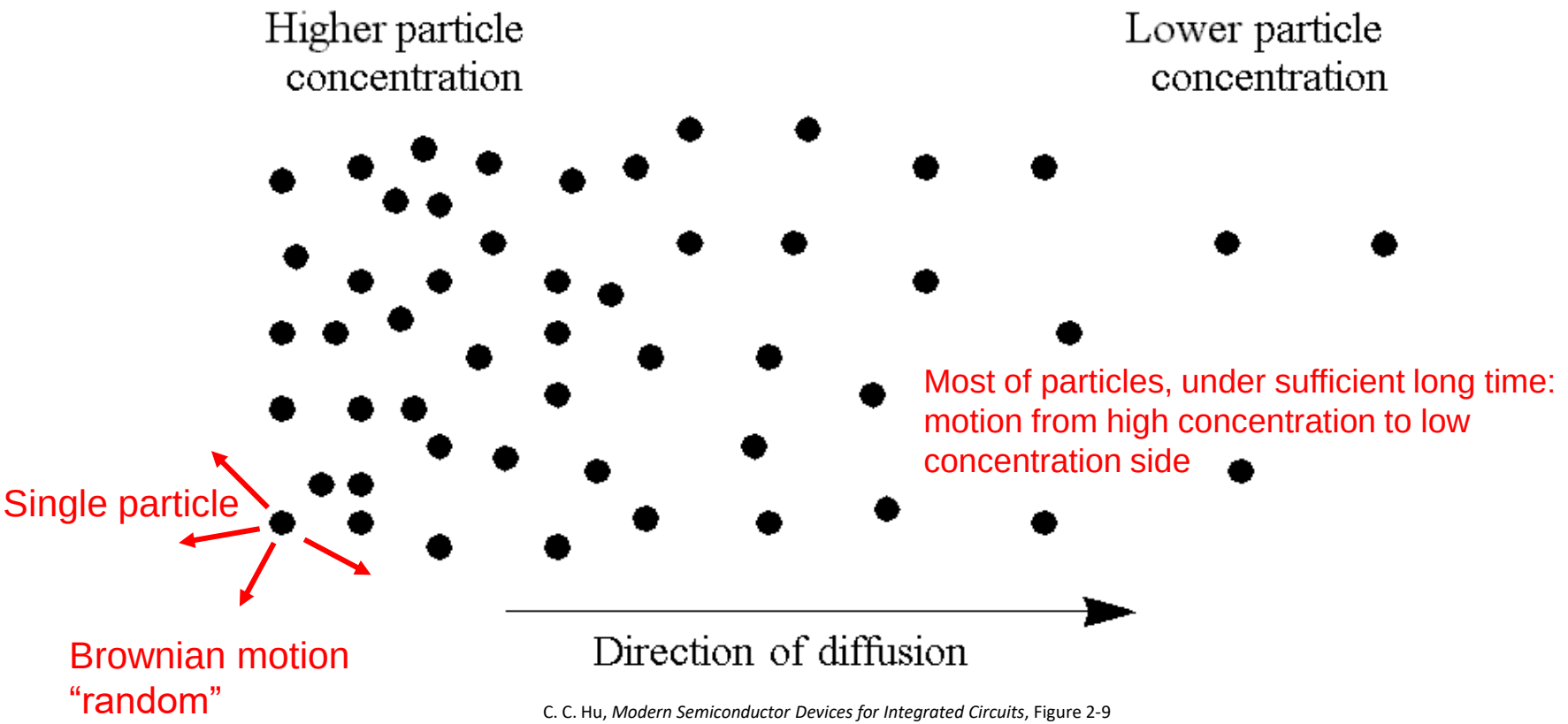
The temperature dependent factor in $\sigma$ (and therefore $\rho$) is $\mu_n$.

From the mobility *vs.* temperature curve for $10^{17}$ cm$^{-3}$, we find that $\mu_n$ decreases from 770 at 300K to 400 at 400K.

Thus, $\rho$ increases by $\frac{770}{400} = 1.93$

# Diffusion

Particles diffuse from regions of higher concentration to regions of lower concentration region, due to random thermal motion.




C. C. Hu, *Modern Semiconductor Devices for Integrated Circuits*, Figure 2-9

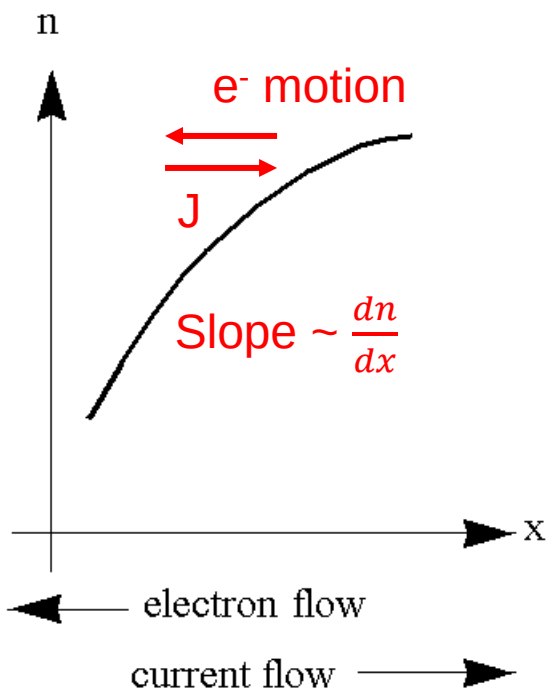
# Diffusion Current

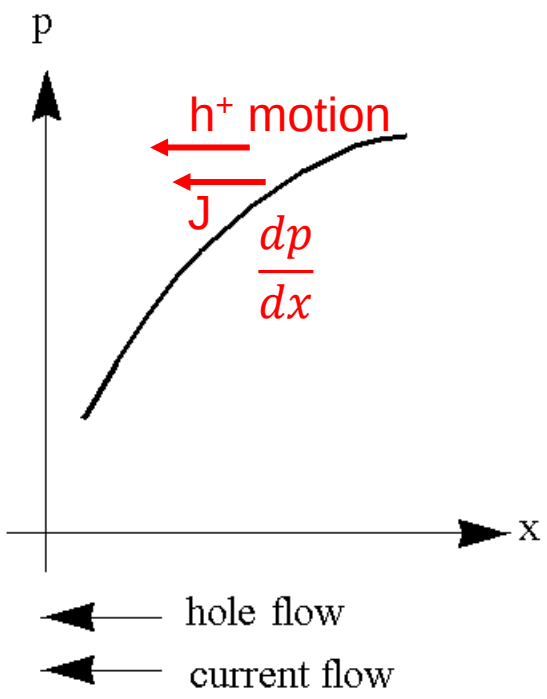
$$J_{\text{n,diff}} = qD_n \frac{dn}{dx}$$

$$\frac{\text{A}}{\text{cm}^2} = \frac{\text{C}}{\text{s}\cdot\text{cm}^2}$$

$$J_{\text{p,diff}} = -qD_p \frac{dp}{dx}$$

$$\frac{\text{C}}{\frac{\text{cm}^{-3}}{\text{cm}}}$$




*D* is the ***diffusion constant***, or ***diffusivity***. Unit: $\frac{\text{cm}^2}{\text{s}}$

# Total Current

$$\boxed{J = J_n + J_p}$$

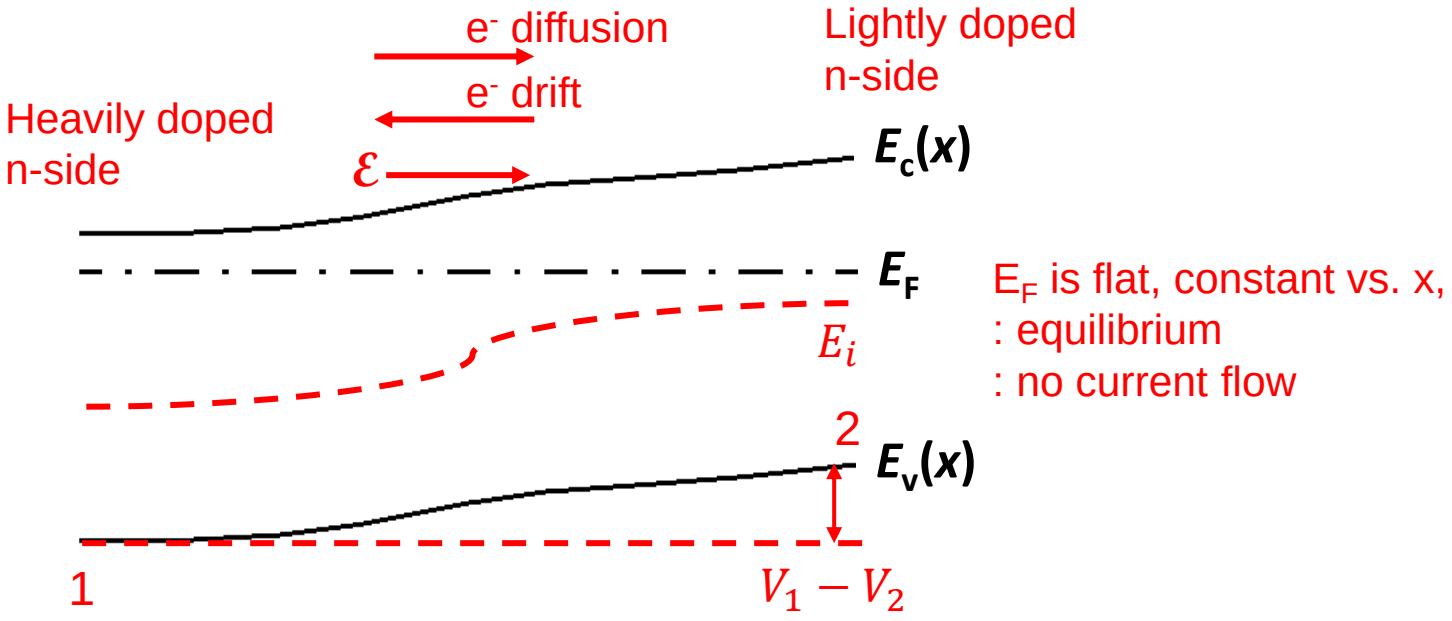
Electron: $J_n = J_{n,drift} + J_{n,diff} = qn\mu_n\mathcal{E} + qD_n\frac{dn}{dx}$

Drift current always along E-field

Hole: $J_p = J_{p,drift} + J_{p,diff} = qp\mu_p\mathcal{E} - qD_p\frac{dp}{dx}$

# Non-Uniformly-Doped Semiconductor

- The position of $E_F$ relative to the band edges is determined by the carrier concentrations, which is determined by the net dopant concentration.
- **In equilibrium $E_F$ is constant**; therefore, the band-edge energies vary with position in a non-uniformly doped semiconductor:

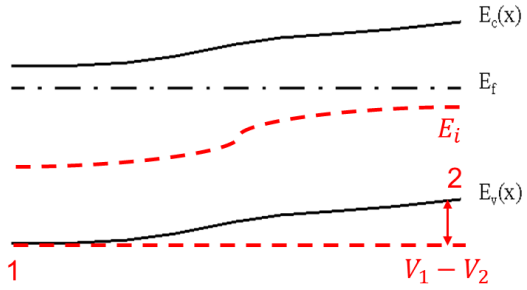


# Potential Difference due to $n(x)$, $p(x)$

- **The ratio of carrier densities at two points depends exponentially on the potential difference between these points:**

$$n = n_i \cdot \exp\left(\frac{E_F - E_i}{kT}\right)$$

$$E_F - E_{i1} = kT \ln\left(\frac{n_1}{n_i}\right) \quad => \quad E_{i1} = E_F - kT \ln\left(\frac{n_1}{n_i}\right)$$




Similarly, $E_{i2} = E_F - kT \ln\left(\frac{n_2}{n_i}\right)$

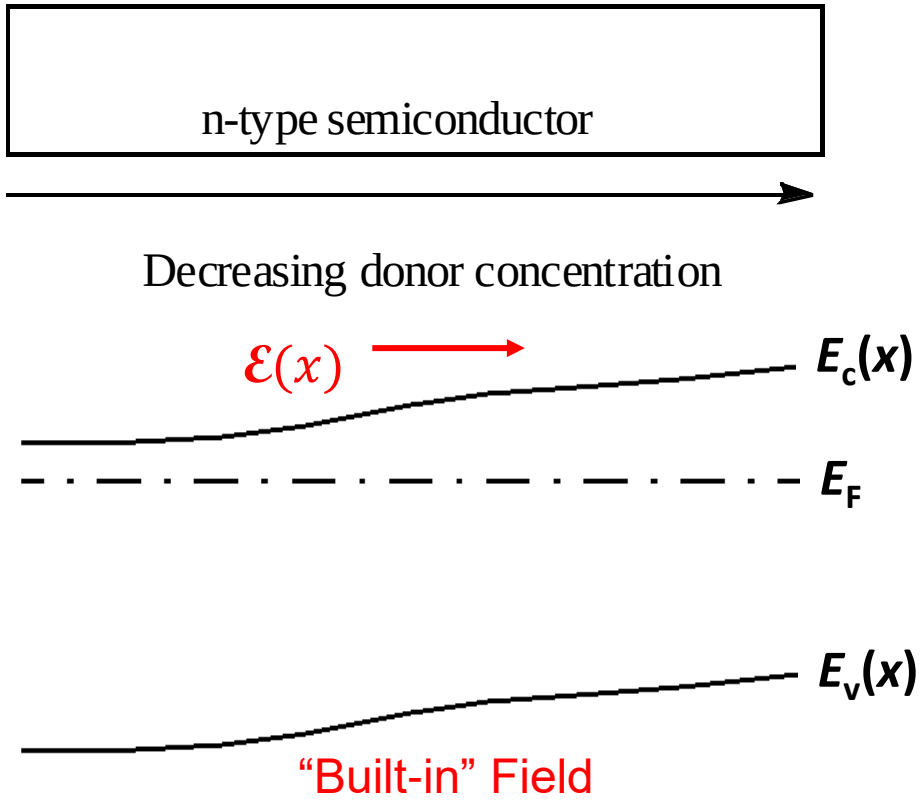
Therefore $E_{i1} - E_{i2} = kT\left[\ln\left(\frac{n_2}{n_i}\right) - \ln\left(\frac{n_1}{n_i}\right)\right] = kT \ln\left(\frac{n_2}{n_1}\right)$

$$V_2 - V_1 = \frac{1}{q}\left(E_{i1} - E_{i2}\right) = \frac{kT}{q} \ln\left(\frac{n_2}{n_1}\right)$$

Potential difference only depends on doping concentration at the 2 ends, regardless of the doping distribution

# Built-In Electric Field due to *n*(*x*), *p*(*x*)

Consider a piece of a non-uniformly doped semiconductor:




$$n = N_c e^{-(E_c - E_F)/kT}$$

$$\frac{dn}{dx} = -\frac{N_c}{kT} e^{-(E_c - E_F)/kT} \frac{dE_c}{dx}$$

$$= -\frac{n}{kT} \frac{dE_c}{dx}$$

$$\mathcal{E} = \frac{1}{q} \cdot \frac{dE_c}{dx}$$

$$= -\frac{n}{kT} q\mathcal{E}$$

$$\mathcal{E} = -\frac{\mathrm{kT}}{q} \cdot \frac{1}{n(x)} \cdot \frac{\partial n(x)}{\partial x}$$

# Einstein Relationship between $D$, $\mu$

- In equilibrium there is no net flow of electrons or holes

$$J_n = 0 \quad \text{and} \quad J_p = 0$$

➔ The drift and diffusion current components must balance each other exactly. (A built-in electric field exists, such that the drift current exactly cancels out the diffusion current due to the concentration gradient.)

$$J_n = qn\mu_n\mathcal{E} + qD_n\frac{dn}{dx} = 0 \qquad J_p = qp\mu_p\mathcal{E} - qD_p\frac{dp}{dx} = 0$$

$$qn\mu_n\mathcal{E} = qn\mu_n\left(-\frac{kT}{q}\cdot\frac{1}{n}\cdot\frac{dn}{dx}\right) = -qD_n\frac{dn}{dx}$$

$$D_n = \frac{kT}{q}\mu_n, \qquad D_p = \frac{kT}{q}\mu_p$$

**The Einstein relationship is valid for a non-degenerate semiconductor, even under non-equilibrium conditions.**

# Example: Diffusion Constant

What is the hole diffusion constant in a sample of silicon with $\mu_p = 410\ \text{cm}^2 / \text{V s}$ ?

Answer:

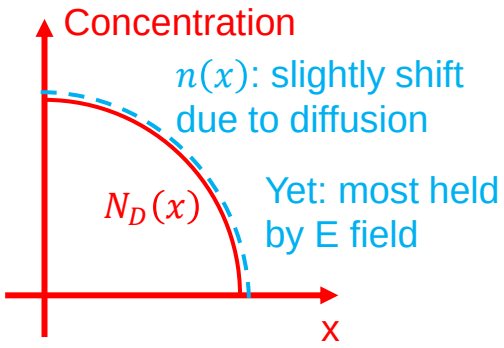
$$D_p = \frac{kT}{q}\mu_p = 26\text{mV} \cdot 410\text{cm}^2/(\text{V} \cdot \text{s}) = 11\text{cm}^2/s$$

Remember: ***kT/q* = 26 mV** at room temperature.

# Quasi-Neutrality Approximation

- If the dopant concentration profile varies gradually with position, then the majority-carrier concentration distribution does not differ much from the dopant concentration distribution.

Strict charge neutrality: $N_{\mathrm{D}}(x) + p(x) = N_{\mathrm{A}}(x) + n(x)$



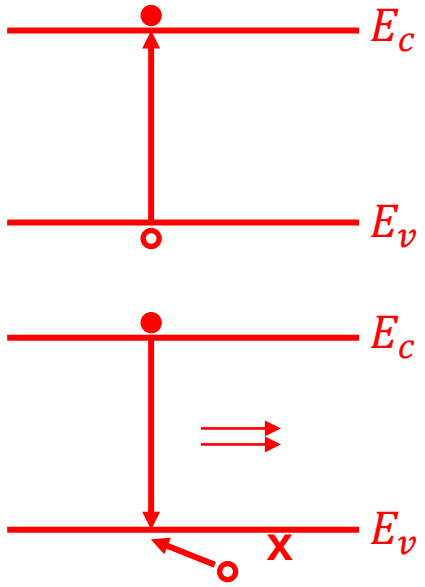

- n-type material: $n(x) \cong N_{\mathrm{D}}(x) - N_{\mathrm{A}}(x)$
- p-type material: $p(x) \cong N_{\mathrm{A}}(x) - N_{\mathrm{D}}(x)$

$$\rightarrow \quad \mathcal{E} = -\frac{kT}{q}\left(\frac{1}{n}\right)\frac{dn}{dx} = -\frac{kT}{q}\left(\frac{1}{N_{\mathrm{D}}}\right)\frac{dN_{\mathrm{D}}}{dx}$$ in n-type material

# Generation and Recombination

Happens in electron-hole pair

- Generation:
  - A process by which electrons & holes are created in pairs.
- Recombination:
  - A process by which electrons and holes are annihilated in pairs.




Disturbance, causing $\Delta J$

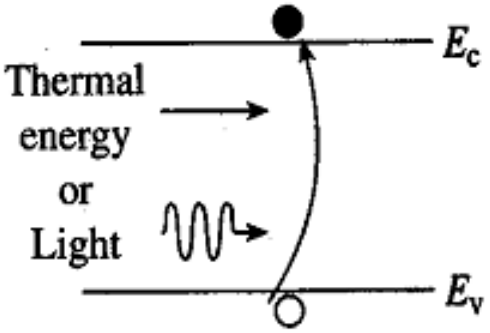
$$J_n = J_{\text{drift}}(n) + J_{\text{diffusion}}\left(\frac{\partial n}{\partial x}\right)$$

- Generation and recombination processes act to change the carrier concentrations, and thereby indirectly affect current flow

# Generation Processes

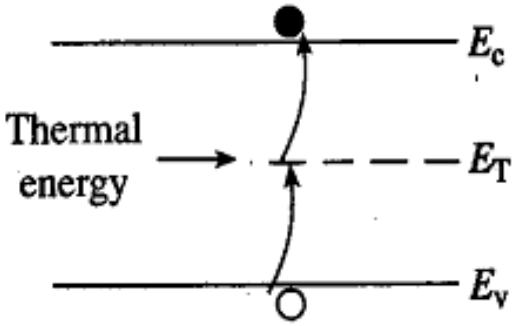
R.F. Pierret, *Semiconductor Fundamentals*, Figure 3.15

## Band-to-Band




## R-G Center

"Trap-assisted"

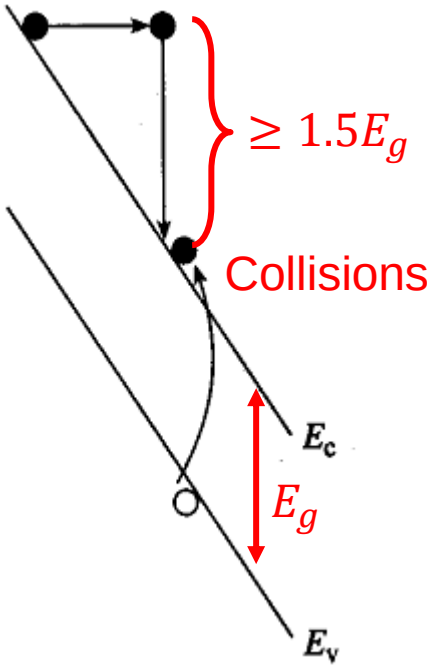



Traps: Au, Cu, Pt, defects in Si

Dominant in Si, Ge

## Impact Ionization

Under high E-field



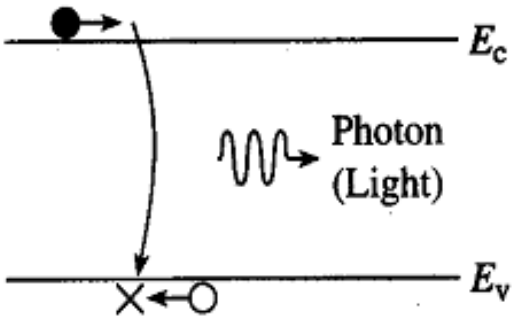
# Recombination Processes

R.F. Pierret, *Semiconductor Fundamentals*, Figure 3.15

Occurs in heavily doped semiconductor

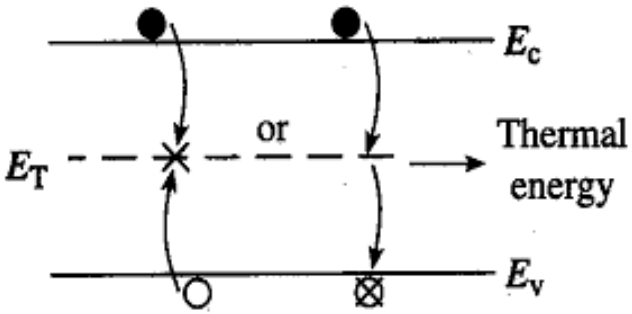
## Direct

GaAs, InP…(III-V)

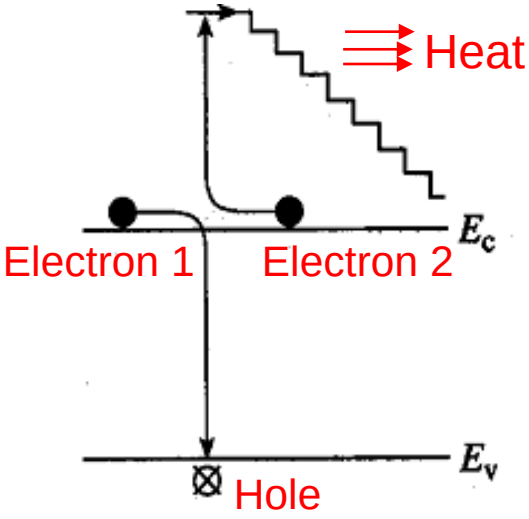



## R-G Center

Si, Ge (Ⅳ)




## Auger




"3-particle" process

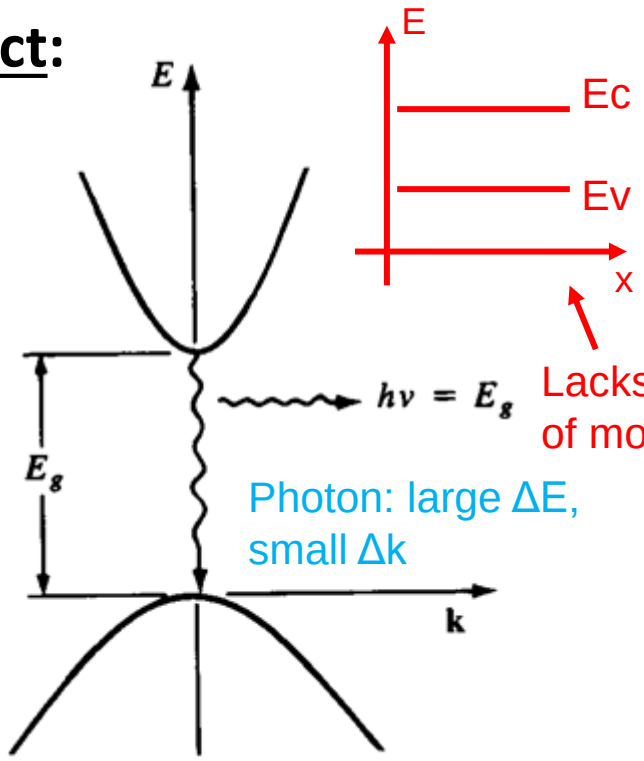
**Recombination in Si is primarily via R-G centers**

# Direct *vs.* Indirect Band Gap Materials

## Energy ($E$) *vs.* momentum ($\boldsymbol{p}=\hbar\boldsymbol{k}$) Diagrams

**Direct:**

Ⅲ-V




Lacks the info of momentum

Photon: large ΔE, small Δk

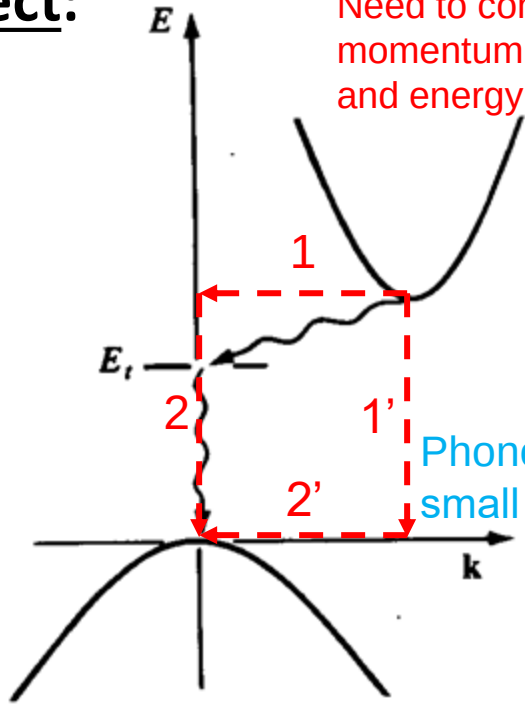
**Little change in momentum is required for recombination**

→ **momentum is conserved by photon emission**

**Indirect:**

Si, Ge

Need to convert both momentum (phonon) and energy (photon)

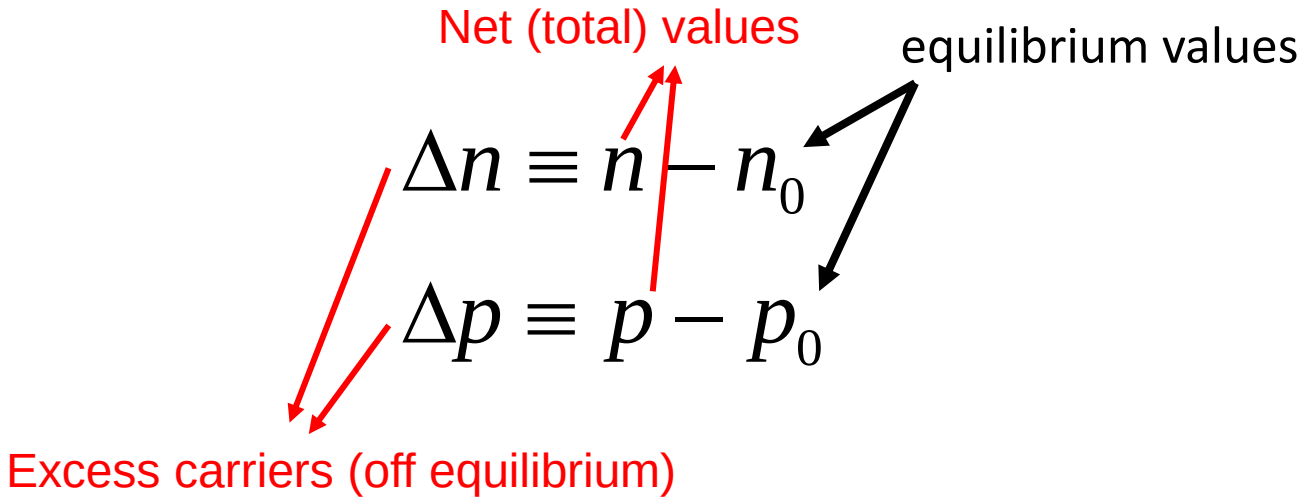



Phonon: large Δk, small ΔE

**Large change in momentum is required for recombination**

→ **momentum is conserved by phonon + photon emission**

# Excess Carrier Concentrations




Charge neutrality condition:

$$\Delta n = \Delta p$$

Electrons and holes R-G in pairs

# “Low-Level Injection”

- Often the disturbance from equilibrium is small, such that **the majority-carrier concentration is not affected significantly**:
  - For an n-type material:

$$|\Delta n| = |\Delta p| << n_0 \quad \text{so} \quad n \cong n_0$$

  - For a p-type material:

$$|\Delta n| = |\Delta p| << p_0 \quad \text{so} \quad p \cong p_0$$

**However, the minority carrier concentration can be significantly affected.**

# Summary

- Electrons and holes can be considered as quasi-classical particles with effective mass *m**
- In the presence of an electric field $\mathcal{E}$, carriers move with **average drift velocity $v_d = \mu\mathcal{E}$ , $\mu$ is the carrier mobility**
  - **Mobility decreases w/ increasing <u>total concentration of ionized dopants</u>**
  - **Mobility is dependent on temperature**
    - decreases w/ increasing *T* if lattice scattering is dominant
    - decreases w/ decreasing *T* if impurity scattering is dominant
- The conductivity ($\sigma$) hence the resistivity ($\rho$) of a semiconductor is dependent on its mobile charge carrier concentrations and mobilities

$$\sigma \equiv qp\mu_p + qn\mu_n \qquad \rho \equiv 1/\sigma$$

# Summary

- **Electron/hole concentration gradient → diffusion**

$$J_{\text{n,diff}} = qD_n \frac{dn}{dx} \qquad J_{\text{p,diff}} = -qD_p \frac{dp}{dx}$$

- **Current flowing in a semiconductor is comprised of drift and diffusion components for electrons and holes**

$$J = J_{\text{n,drift}} + J_{\text{n,diff}} + J_{\text{p,drift}} + J_{\text{p,diff}}$$

In equilibrium $J_n = J_{\text{n,drift}} + J_{\text{n,diff}} = 0$ and $J_p = J_{\text{p,drift}} + J_{\text{p,diff}} = 0$

- **The characteristic constants of drift and diffusion are related:**

$$\frac{D}{\mu} = \frac{kT}{q}$$

# Acknowledgement


- Modern Semiconductor Devices for Integrated Circuits by C. Hu (Prentice Hall, 2009)
- Semiconductor Device Fundamentals by R. F. Pierret (Addison Wesley, 1996)
- Professor Tsu-Jae King Liu EE130/230A notes